WORKED SOLUTION

STEP I Specimen paper, Question 11

Mechanics • Friction and Rolling Contact

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q11

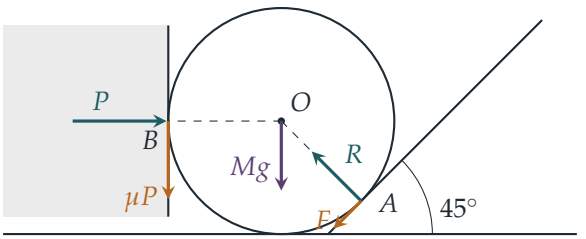
The key idea

At the moment the cylinder is about to move, the floor stops pushing, so only three contacts matter: the weight, the vertical face and the ramp. "Roll" means the cylinder does not slip on the ramp, so it starts to turn about its point of contact with the ramp. Work out how the point touching the vertical face then moves, and you see that the cylinder has to slide against the face. So that is where friction is limiting. A moment equation about the centre then gives the ramp friction with no extra work.

Solution

Model. The cylinder is a uniform rigid body of radius r, and we treat the problem in the vertical cross-section through its centre O. The equipment's face is vertical and moves horizontally. Let B be the point of contact with the face, which is level with O, and A the point of contact with the ramp. Since the ramp is at 45°, OA makes 45° with the downward vertical. Take x to the right (towards the ramp) and y upwards, with O as origin, so

$$B = (-r, 0), \qquad A = \tfrac{r}{\sqrt{2}}(1, -1).$$



Which friction is limiting

When the cylinder is on the point of moving, it is about to leave the floor, so the floor's normal reaction is zero, and with it the floor's friction. To roll up the ramp, the cylinder

must not slip at A, so its motion starts as a rotation about A, with the centre moving up the slope in the direction $(1,1)$. That rotation is clockwise. With angular speed ω, a point X of the cylinder has velocity ω times $X-A$ turned through 90° clockwise, that is $\omega\,(y,\,-x)$ applied to $X-A=(x,\,y)$. For $X=B$,

$$B-A=r\Big(-1-\tfrac{1}{\sqrt{2}},\,\tfrac{1}{\sqrt{2}}\Big)\quad\Longrightarrow\quad \mathbf{v}_B=\omega r\Big(\tfrac{1}{\sqrt{2}},\,1+\tfrac{1}{\sqrt{2}}\Big).$$

The face moves only horizontally, but B has an upward velocity component, so B slides up the face. The cylinder cannot roll on both surfaces at once, so it must slip on the vertical face, and **the friction is limiting on the vertical face**. Friction opposes the sliding, so it acts *down* on the cylinder with magnitude μP. At the ramp there is no slipping, so the ramp friction F is whatever equilibrium requires, subject only to $F\le \nu R$. (We write ν for the ramp's coefficient of friction, printed as v in the question.)

Ask yourself Why can't friction be limiting at the ramp instead?

If the cylinder slipped at A, it would be sliding up the ramp, not rolling. Rolling means no slip at A. The calculation above shows that rolling forces slip at B, so B is the contact where friction reaches μ times the normal reaction.

Finding P

Let R be the normal reaction of the ramp, along AO (direction $\frac{1}{\sqrt{2}}(-1,1)$), and let F be the ramp friction, taken as positive *down* the slope (direction $\frac{1}{\sqrt{2}}(-1,-1)$).

Moments about O. The weight, P and R all pass through O. The friction μP at B acts at distance r and turns the cylinder anticlockwise. The friction F at A is tangential, so it also acts at distance r, and when it points down the slope it turns the cylinder clockwise. Hence

$$\mu P\,r=Fr\quad\Longrightarrow\quad F=\mu P.$$

So the ramp friction really does act down the slope, and it has the same size as the friction at the face.

Resolving horizontally:

$$P-\frac{R}{\sqrt{2}}-\frac{\mu P}{\sqrt{2}}=0\quad\Longrightarrow\quad R=\sqrt{2}\,P-\mu P=P(\sqrt{2}-\mu).\tag{1}$$

Resolving vertically:

$$\frac{R}{\sqrt{2}}-\frac{\mu P}{\sqrt{2}}-\mu P-Mg=0.\tag{2}$$

Substituting $R/\sqrt{2}=P-\mu P/\sqrt{2}$ from (1) into (2),

$$\begin{aligned}P-\frac{\mu P}{\sqrt{2}}-\frac{\mu P}{\sqrt{2}}-\mu P&=Mg\\ P(1-\mu-\sqrt{2}\,\mu)&=Mg,\end{aligned}$$

using $2/\sqrt{2} = \sqrt{2}$. Therefore

$$P = \frac{Mg}{1 - \mu(1 + \sqrt{2})},$$

as required.

The conditions on μ and ν

On μ. P is the magnitude of a push, and $Mg > 0$, so we need $1 - \mu(1 + \sqrt{2}) > 0$, that is

$$\mu < \frac{1}{1 + \sqrt{2}} = \frac{\sqrt{2} - 1}{(\sqrt{2} + 1)(\sqrt{2} - 1)} = \sqrt{2} - 1.$$

If $\mu \geq \sqrt{2} - 1$, no push, however large, will roll the cylinder up the ramp: the extra downward friction at the face grows with P at least as fast as the lift does, and the cylinder jams.

On ν. The cylinder does not slip on the ramp, so $F \leq \nu R$. Using $F = \mu P$ and (1), and noting that $R > 0$ because $\mu < \sqrt{2} - 1 < \sqrt{2}$,

$$\mu P \leq \nu P(\sqrt{2} - \mu) \quad \Longleftrightarrow \quad \nu \geq \frac{\mu}{\sqrt{2} - \mu},$$

where we divided by $P(\sqrt{2} - \mu) > 0$. This completes the question.

Checks

- *Smooth face*, $\mu = 0$: then $P = Mg$ and no ramp friction is needed. This matches a frictionless cylinder wedged between a vertical wall and a 45° ramp: the ramp's reaction has equal horizontal and vertical components, so it must balance P and Mg equally.
- *Numbers*: with $\mu = 0.2$, $P \approx 1.934\,Mg$ and we need $\nu \geq 0.165$. Substituting these values back, the horizontal force, vertical force and moment about O all balance (to rounding error), for random M and μ too.
- As $\mu \to \sqrt{2} - 1 \approx 0.414$, $P \to \infty$, which is consistent with the jamming described above.

Common slips

- Making the friction limiting at the ramp. "Roll" rules out slipping there.
- Leaving the floor reaction in. At the point of rolling up, the cylinder is about to lift off, so the floor's reaction is zero.
- Getting the direction of the face friction wrong. B moves *up* relative to the face, so friction on the cylinder acts *down*. This is why P exceeds Mg.

Beyond STEP

The key step, deciding which contact slips by looking at the instantaneous centre of rotation, is the start of rigid-body kinematics. In a first-year course it leads on to the general rule $\mathbf{v}_X = \mathbf{v}_O + \boldsymbol{\omega} \times (X - O)$ and to problems where several rough contacts compete. The "jamming" at $\mu \geq \sqrt{2} - 1$, where pushing harder never helps, is a self-locking effect that engineers use deliberately in wedges and clamps. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body motion and friction with this kind of care.

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